Project Report: Improving Bank Telemarketing Conversion

Sector: Finance / Retail Banking
Project Team: Velocity Vortex
Institute: Newton School of Technology

1. Executive Summary

This report presents a comprehensive analysis of a direct marketing campaign conducted by a Portuguese banking institution between May 2008 and November 2010. The primary objective was to identify the key drivers of term-deposit subscriptions and provide actionable recommendations to enhance campaign efficiency.

Our analysis, powered by a robust Python-based ETL pipeline and statistical framework, reveals that the overall conversion rate stands at a modest **11.27%**. However, specific segments show significantly higher performance. Most notably, customers who had a successful outcome in previous campaigns converted at **65.11%**, and cellular-based outreach outperformed traditional telephone lines by nearly **3:1**.

Key findings indicate a clear "campaign fatigue" effect, where conversion rates drop sharply after the third contact. By refocusing efforts on high-performing segments—such as students, retired individuals, and previous successful responders—and optimizing contact frequency, the bank can substantially reduce wasted outreach effort and improve its return on marketing investment.

2. Sector Context and Problem Statement

2.1 Sector Context

The period between 2008 and 2010 was marked by the global financial crisis, which severely impacted the Portuguese banking sector. During this time, banks faced restricted access to international wholesale funding markets, making domestic deposits a critical source of liquidity and stability. Term deposits, in particular, provided banks with a predictable capital base to manage their balance sheets amidst market volatility.

2.2 Problem Statement

Despite the strategic importance of term deposits, the bank's current "broad-brush" telemarketing approach is inefficient. With an overall subscription rate of only 11.27%, the majority of outreach efforts result in no conversion, leading to high operational costs and potential customer annoyance.

The core business challenge is to transition from an unstructured outreach model to a **data-driven targeting strategy**. Specifically, the bank needs to answer:

- Which customer demographics (age, job, education) are most likely to subscribe?
- What campaign attributes (contact channel, timing, frequency) correlate with success?
- How does previous interaction history influence current campaign outcomes?

---

3. Data Description

3.1 Data Source and Structure

The analysis is based on the **UCI Bank Marketing Dataset** (`bank-additional-full.csv`), which contains 41,188 records of phone-based marketing contacts.

Feature Category	Description	Examples
Demographics	Client personal data	Age, Job, Marital Status, Education
Financials	Credit and loan status	Default, Housing Loan, Personal Loan
Campaign	Current contact details	Contact Type, Month, Day, Duration
Social/Economic	External context	Employment Variation Rate, CPI, Euribor 3m
History	Previous interactions	Pdays, Previous, Poutcome
Target	Outcome variable	`y` (Subscribed: Yes/No)

3.2 Size and Limitations

- **Size:** 41,188 rows and 21 columns.
- **Limitations:**
  - **Unknown Values:** Several categorical fields (Job, Education, etc.) contain "unknown" labels, which were treated as missing data during analysis.
  - **Duration Bias:** The `duration` field (call length) is only known after the call is completed. While it is a strong predictor of success, it cannot be used for pre-call targeting.
  - **Temporal Scope:** The data covers a specific economic period (2008-2010); findings may need recalibration for current market conditions.

---

4. Data Cleaning and ETL Methodology

We developed a standardized Python pipeline (`etl_pipeline.py`) to transform raw data into an analysis-ready format.

4.1 Processing Steps

1. **Normalization:** Standardized column names to lowercase and replaced special characters with underscores.
2. **Missing Value Handling:** Identified "unknown" strings in categorical columns and replaced them with `NaN` to allow for proper statistical handling.

3. **Feature Engineering:**
    - **Age Grouping:** Binned ages into six categories (18-24, 25-34, …, 65+).
    - **Campaign Banding:** Grouped the number of contacts into bands (1, 2, 3, 4-5, 6-10, 11+).
    - **Recency Transformation:** Converted `pdays` (999 for never contacted) into a boolean `contacted_before` flag.
    - **Temporal Mapping:** Mapped month and day abbreviations to full names and established a chronological sort order.
4. **Export:** Generated a "Tableau-ready" CSV and several pre-aggregated KPI tables for dashboarding.

---

5. KPI and Metric Framework

To measure campaign success and guide decision-making, we established the following framework:

Metric	Definition	Current Value
Subscription Rate	% of contacts resulting in a "Yes"	11.27%
Contact Efficiency	Conversion rate by channel (Cellular vs. Tel)	14.7% vs. 5.2%
Retention Signal	Conversion rate for previous "Success" clients	65.11%
Campaign Fatigue	Drop in conversion after 'N' contacts	-76% (1 vs. 11+ contacts)
Average Duration	Mean call length (proxy for engagement)	258 seconds

---

6. EDA with Visualizations and Insights

6.1 Contact Channel Performance

Insight: Cellular contacts are significantly more effective than landline (telephone) contacts.

- **Cellular:** 14.74% Conversion
- **Telephone:** 5.23% Conversion *Strategy:* The bank should prioritize mobile-first outreach.

6.2 The Power of Previous Success

Insight: A "Success" in a previous campaign is the single strongest predictor of future subscription.

- **Previous Success:** 65.11% Conversion
- **No Previous Contact:** 8.83% Conversion *Strategy:* Create a "Gold List" of previous subscribers for immediate follow-up in every new campaign.

6.3 Demographic Sweet Spots

Insight: The "U-shaped" age response.

- The highest conversion rates are found in the **18-24** (Student) and **65+** (Retired) age groups.
- The middle-aged workforce (30-50) shows the lowest relative interest.

6.4 Campaign Fatigue

Insight: Repeated attempts yield diminishing returns.

- **1 Contact:** 13.04% Conversion
- **4-5 Contacts:** 6.5% Conversion
- **11+ Contacts:** 3.11% Conversion *Strategy:* Implement a "Hard Cap" at 5 contacts to prevent resource wastage.

---

7. Statistical Analysis Results

We performed rigorous testing to validate our findings (Alpha = 0.05).

Test Type	Variables	Result (p-value)	Conclusion
Chi-Square	Contact Type vs. Subscribed	< 0.001	Strong association; channel matters.
Chi-Square	Prev. Outcome vs. Subscribed	< 0.001	Strong association; history is key.
Chi-Square	Job Type vs. Subscribed	< 0.001	Significant difference across segments.
T-Test	Duration (Sub vs. Non-Sub)	< 0.001	Subscribers stay on the line longer (553s vs 221s).

---

8. Dashboard Design

The **Bank Campaign Performance Dashboard** was designed in Tableau to provide real-time visibility into these metrics.

8.1 Key Components

- **Executive Summary Tiles:** High-level view of Total Records, Total Subscribers, and Overall Conversion %.
- **Performance Heatmaps:** Subscription rates by Month and Day of the Week to optimize timing.
- **Segment Deep-Dive:** Interactive filters for Job, Education, and Age Group.
- **Campaign Fatigue Funnel:** Visualizing the drop-off in conversion as contact frequency increases.

8.2 Screenshot Explanation

(Note: Screenshots are stored in the `tableau/screenshots/` directory) The dashboard allows managers to filter by "Previous Outcome." When "Success" is selected, the entire dashboard shifts to show the characteristics of this high-value segment, enabling precise targeting.

---

9. Key Insights (Decision Language)

1. **Stop Landline Primacy:** Telephone contacts convert at less than half the rate of cellular; shift 80%+ of volume to mobile.
2. **Prioritize the "Success" Segment:** Customers who said "Yes" before are 7x more likely to say "Yes" again than the average prospect.
3. **The Rule of Five:** Do not exceed 5 contact attempts for a single campaign; the cost-per-acquisition becomes prohibitive beyond this point.
4. **Target the Life Stages:** Focus messaging on "Students" (early savers) and "Retired" (wealth preservation) who show the highest affinity for term deposits.
5. **Seasonal Optimization:** March, September, and October show peak conversion rates; concentrate major campaign launches in these windows.
6. **Engagement Proxy:** While duration cannot be targeted, calls shorter than 180 seconds rarely result in conversion. Use this as a training benchmark for sales agents.
7. **Economic Sensitivity:** Subscription rates move inversely with the Euribor 3-month rate; as rates rise, term deposits become more attractive.
8. **Avoid "Unknowns":** Clients with missing job/education data convert at lower rates; prioritize records with complete profiles.

---

10. Actionable Business Recommendations

1. **Implement a Tiered Outreach Model:**
    - **Tier 1 (High Priority):** Previous Successes + Students/Retired + Cellular.

    - **Tier 2 (Medium Priority):** New prospects in high-conversion months (Mar/Sep/Oct).
    - **Tier 3 (Low Priority):** Middle-aged workforce + Landline contacts.

2. **Enforce a Contact Frequency Cap:** Program the dialer to automatically flag and move records to a "Cool-off" list after 5 unsuccessful attempts in a single campaign.
3. **Incentivize Cellular Data Collection:** Offer a small interest rate "top-up" or fee waiver for customers who provide and verify a mobile number, as this channel is the primary driver of conversion.

---

11. Impact Estimation, Limitations, and Future Scope

11.1 Impact Estimation

By shifting just 20% of telephone volume to cellular and capping contacts at 5, we estimate a **15-20% increase in overall conversion rate** and a **10-15% reduction in operational costs** related to wasted outreach.

11.2 Limitations

- **External Factors:** The model does not account for competitor interest rates during the 2008-2010 period.
- **Data Freshness:** Consumer behavior regarding telemarketing has changed significantly since 2010 (e.g., rise of spam filters).

11.3 Future Scope

- **Predictive Modeling:** Transition from descriptive analytics to a machine learning model (e.g., Random Forest) to predict individual subscription probability.
- **Sentiment Analysis:** Integrate call transcripts to analyze the "voice of the customer" and refine sales scripts.

---

12. Contribution Matrix

This matrix reflects the actual work distribution as recorded in the project repository and GitHub history.

Member	Responsibility Area	Key Deliverables
Madhur Rathod	Project Lead	Repo coordination, QA, final integration, README
Aaditya Mohan Samadhiya	Data Engineer	ETL pipeline (`scripts/etl_pipeline.py`), Cleaning (`02_cleaning.ipynb`)
Ambuj Vashistha	Lead Analyst	EDA (`03_eda.ipynb`), Statistical Testing (`04_statistical_analysis.ipynb`)
Himanshu Gulhane	BI & Storytelling	Tableau Dashboard, Project Report, Presentation Deck

---